

Group Assignment 4 : Ideation

Team Fitech

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Persona

Persona 1: Kim Minji (23)

- Background: A student and dedicated gym-goer aiming for consistent muscle growth.
- Motivation: Driven by the need to make the most of her limited time after classes and assignments, Minji is motivated by measurable physical progress such as progressive overload. She sees gym time as both a physical challenge and a mental reset from academic stress, and wants to stay fully focused on pushing her limits without letting the administrative "paperwork" of device management interrupt her workout.
- Behavioral Patterns: Visits the gym 4–5 times a week. She strictly follows a pre-planned routine and adheres to precise rest intervals to optimize muscle growth .
- Goal: To maximize workout efficiency by maintaining a continuous, uninterrupted "Flow State."
- Key Characteristics: She frequently wears lifting straps, making smartphone operation physically difficult. Sweaty hands often cause the touchscreen to malfunction. She finds it frustrating and "immersion-breaking" to have to take off her gear or wipe her hands just to log a set or check the timer.

Persona 2: Shin Taehyeong (28)

- Background: A busy office worker who recently started going to the gym to improve his health and relieve stress. Because of his unpredictable schedule, he usually works out during peak hours. As a beginner, he is still unfamiliar with many machines and does not feel confident planning workouts on his own.
- Motivation: He wants to build a consistent workout habit and finish gym sessions without getting distracted or losing momentum.
- Behavioral patterns: Goes to the gym 2–3 times a week, depending on his work schedule. He Often works out during peak hours when the gym is crowded and machines are frequently occupied. He Often gets distracted after checking his phone and loses focus during workouts
- Goal: To complete workouts confidently and continuously without relying on his phone so much.
- Key characteristics: He is a beginner user who does not know how to use some machines well. He gets easily distracted when using his phone during workouts. He gets confused when equipment is occupied and loses momentum easily.

Problem Statement

[Gym users] need to [seamlessly manage their workout routines and find alternatives without visually engaging with a screen] because [current smartphone interfaces impose physical friction on heavy lifters and trigger digital distractions and cognitive overload for beginners.]

Six How Might We (HMW) Questions

For Kim Minji (Expert / Focus: Physical Constraints)

1. **(Remove the bad)** HMW eliminate the need for manual touch interaction so Minji doesn't have to remove her lifting gloves or wipe her hands to log a set?
2. **(Amp up the good)** HMW leverage audio-delivered performance insights and historical growth data to guide Minji's progressive overload without disrupting her physical momentum?

3. **(Create an analogy)** HMW make the mobile app function as a **"Hands-free Assistant"** that handles all "paperwork" (data entry) while she focuses entirely on the heavy lifting?

For Shin Taehyeong (Beginner / Focus: Digital Distraction & Occupied Equipment)

4. **(Question an assumption)** HMW provide exercise guidance and alternative options without requiring Taehyeong to unlock his phone and face the temptation of social media?
5. **(Explore the opposite)** HMW design a workout experience where the app "talks" to Taehyeong through his routine without interaction to screen, preventing mindless scrolling during rest periods?
6. **(Remove the bad)** HMW reduce the anxiety of **"what to do next"** when equipment is busy by proactively suggesting alternatives through audio, keeping Taehyeong from feeling lost or looking distracted?

Solution Ideation (12 to 3)

For Kim Minji (Expert)

- **HMW 1: Eliminate manual touch (gloves/sweat)**

1. **Proactive Audio Routine Briefing:** At the end of a rest interval, the app automatically dictates the next set: *"Next set: 100kg, 5 reps. Prepare to start."* This allows Minji to grip the bar immediately without ever looking at her screen.
2. **Smart-Strap Motion Detection:** Using the smartphone or smartwatch accelerometer to automatically detect when a set is finished and log it based on the pre-planned weight.

- **HMW 2: Provide Data-Driven Performance Insights and Growth Strategies**

3. **Historical Performance Briefing:** Before workout, the app provides a brief audio summary of her recent performance trends and achievements compared to previous sessions to sustain motivation.
4. **AI Progressive Overload Proposer:** The app analyzes her historical data to verbally suggest specific weight or rep increases for the current session, providing a roadmap for continuous growth without manual tracking.

- **HMW 3: Hands-free Assistant "Paperwork"**

5. **Post-Workout Voice Summary:** At the end of the session, Minji provides a brief voice memo of how she felt, which the app automatically converts into a text-based progress diary.
6. **Earbud Gesture Control:** Using simple head movements (e.g., a double-nod) or earbud taps to confirm "Set Complete" and trigger the rest timer instantly.

For Shin Taehyeong (Beginner)

- **HMW 4: Guidance without unlocking (Distraction)**

7. **Lock-screen "Focus Widget":** A simplified, non-interactive display that shows only the current exercise and next steps on the lock screen, preventing the temptation to open SNS.
8. **Proactive Audio Form-Check:** The app "speaks" the top three posture tips (e.g., "Keep your heels down") as soon as it detects the start of a specific exercise.

- **HMW 5: Talk-through routine (No screen)**

9. **Zen-Mode Audio Sequence:** A "Screen-Locked" mode where the app acts like a podcast, guiding Taehyeong through the entire routine—including transition times—so he never reaches for his pocket.
10. **Notification "Gym Shield":** An automatic feature that suppresses all non-emergency notifications (like Instagram or KakaoTalk) the moment the workout starts.

- **HMW 6: Proactive alternatives (Occupancy anxiety)**

11. **Audio Goal Planner:** Pre-workout, Taehyeong selects his workout goal for the day. "I want to do a 40-minute chest workout," and the AI immediately dictates a customized plan based on his level and real-time gym occupancy.

12. **"Skip-to-Substitute" Audio Command:** If Taehyeong sees his next machine is occupied, he simply says "Find alternative." The AI identifies his current location and immediately suggests a different exercise that targets the **same muscle group** using available equipment nearby.

Top 3 Selected Solutions & Development

1. The "Invisible Coach" (Voice-First + Hands-Free Workout Support)

A voice-first, hands-free workout assistant that minimizes screen interaction during exercise. Through wireless earbuds, it provides real-time audio guidance such as next exercise, rest time, and target reps or weight, while automatically recording workout data in the background. Users can control the session with simple earbud taps like skip, pause, and resume. This helps experienced users maintain flow and reduces distraction and uncertainty for beginners.

2. AI-Powered Dynamic Routine Planner

An adaptive workout planner that reduces manual decision-making before and during exercise. At the start of the session, it creates a routine based on the user's goals and available time. During the workout, if equipment is occupied or the user wants to skip, the system suggests alternative exercises for the same muscle group and reorganizes the routine in real time. This helps users continue their workout without stopping to re-plan on their phone.

3. Automated Post-Workout Analytics & Progression Tracking

A post-workout mobile app feature that turns automatically collected workout data into clear progress insights. After the session, users can review summaries such as total volume, consistency, and progress over time. It supports experienced users who want accurate performance tracking and gives beginners encouraging feedback through easy-to-understand visual reports. This reduces manual logging effort while helping users stay motivated for future workouts.

Three Scenarios

Scenario 1: Minji uses the "invisible coach"

After class, Minji arrives at the campus gym for her regular upper-body workout. She already has a clear routine in mind and wants to move through it efficiently with strict rest intervals. During her workout, she wears lifting straps and her hands become sweaty, so using her phone to log sets or check the timer feels annoying and immersion-breaking. Instead of reaching for her phone, Minji hears a short audio cue through her wireless earbuds: "Set complete. Rest for 90 seconds." While she recovers, the system automatically records the set in the background and then says, "Next: shoulder press, 10 reps." This lets her continue the workout without stopping to unlock her phone or remove her gear.

As the session continues, Minji stays focused on lifting while the app guides her through each step with simple, timely audio prompts. When she needs a little more time before the next set, she taps her earbud once to pause and then resumes when ready. Because the system gives her rest timing, next-exercise guidance, and automatic logging without requiring screen interaction, Minji avoids both physical friction and unnecessary distraction. By the end of the workout, she feels that the session stayed smooth, controlled, and uninterrupted, which helps her maintain the flow state she values during exercise.

Scenario2 : Taehyeong uses the AI-Powered Dynamic Routine Planner

Taehyeong enters the gym during the peak evening rush, feeling his usual wave of anxiety as he sees almost every major machine occupied. Instead of pulling out his phone to check his notes—a move that usually leads to twenty minutes of mindless scrolling and "looking clueless"—he simply taps his earbuds and says, "I have 40 minutes for a Chest and Shoulder routine today." The AI Dynamic Planner immediately analyzes his fitness level and the gym's current congestion, responding: *"Understood. I've designed a high-efficiency 5-exercise circuit. Let's start with the Incline Dumbbell Press. Head toward the free-weight section."* Taehyeong moves with purpose, feeling like a focused regular rather than a confused beginner.

Upon reaching the dumbbell area, he finds the adjustable benches are all taken. Rather than standing around awkwardly or retreating to his phone's SNS feed, he stays in the zone and whispers, "Find an alternative." Within

seconds, the AI performs Dynamic Re-sequencing and suggests: *"The benches are currently busy. Let's swap to the Hammer Strength Chest Press at Station 3; it's currently open. It targets the same upper chest muscles and will keep your momentum going."* He follows the audio directions and starts his set immediately. By the end of his session, Taehyeong realizes he hasn't looked at his phone screen once, successfully avoiding digital distractions while navigating the crowded environment with total confidence.

Scenario3 : Minji uses Automated Post-Workout Analytics & Progression Tracking

As Minji finishes her final set and heads to the locker room, she no longer feels the "administrative dread" of having to recall and manually input every weight and rep she lifted while exhausted. The moment she closes her workout session via a simple voice command, the Automated Analytics engine processes the data gathered by her sensors and voice confirmations. While cooling down, she opens the app to find a clean, high-contrast dashboard. It doesn't just show a list of numbers; it highlights a "Progressive Overload Milestone," visually charting her 5% increase in squat volume over the last seven days. Seeing her hard work translated into a precise upward trend gives her a sense of professional satisfaction, allowing her to leave the gym with a clear roadmap for her next session without having touched her phone during the actual lift.